

Native determinant returns and finite heat certificates

Consecutive native Gram determinants

$$r_L = d_{L+1}/d_L = (1 + \beta_L)^{-1}$$

$$\varepsilon_L^2 = a_L^2(1 - r_{L-1})(1 - r_L)/r_L$$

$$\Theta \leq \varepsilon_L$$

$$-\log(r_{L-1}r_L) \geq 2\text{arsinh}(\Theta/a_L)$$

Sharp for the stated one-scalar constraint.

$T(t) = A - tQ$, $Q^2 = 0$, original metric G

$$\mathcal{H}_\tau = \text{tr} e^{-\tau T^2} \text{ (holomorphic)}$$

$$\mathcal{P}_\tau = \text{tr} e^{-\tau T^{\dagger_6} T} \text{ (positive)}$$

$$a_1 = \frac{1}{2} \|T - T^{\dagger_6}\|_{\text{HS}, G}^2$$

$$\frac{4}{7} \tau a_1 \leq \Re \mathcal{H}_\tau - \mathcal{P}_\tau \leq \frac{10}{7} \tau a_1 \text{ if } \tau R^2 \leq 1/8$$

The two heat functions are not identified.

$$\mathcal{R}_q = \log \frac{d_{q-1}d_q}{d_{2q-1}d_{2q}} \geq 2 \sum_{L=q}^{2q-1} \text{arsinh}(\Theta/a_L)$$

$$\text{finite stopping: } |\Delta_\tau - \Delta_{\tau, m}| \leq \tau a_1 \frac{(2m+1)x^m}{m! \{1 - 3x/(m+1)\}}$$

The determinant return and heat gap are necessary finite-dimensional constraints;
neither is the whole assembled arithmetic action or a global endpoint theorem.